Test / repaired: Cantera versus NN | paired short and long

Gamma chosen on long-run validation by completion count, then temperature NRMSE. Each pair uses the SAME gamma and common completed test ICs.
NRMSE = RMSE(Cantera, NN) / RMS(Cantera); dimensionless ratio, median across the listed ICs. Lower is better. Green/red: long improved/worsened.

	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2
ANN, g=0.1 | short	0.03213	0.1416	1.261	1.011	0.17	0.9897	0.08139	1.021	1.212	0
ANN, g=0.1 | long	0.01375	0.1136	0.5546	3.712	0.2075	0.9104	0.08083	0.9992	1.413	0
P-only, g=0.2 | short	0.07271	0.3681	1	0.9614	0.2021	0.8568	0.2082	0.987	1.008	0
P-only, g=0.2 | long	0.002405	0.0241	0.1811	0.3249	0.03916	0.09708	0.01918	0.6367	0.765	0
ANN + P-only, g=0.2 | short	0.03521	0.2634	1	0.6768	0.1702	0.6295	0.09243	0.975	0.9979	0
ANN + P-only, g=0.2 | long	0.001663	0.01999	0.1101	0.2247	0.03451	0.09172	0.01225	0.4212	0.5234	0
T/P, g=0.3 | short	0.396	0.3412	0.9785	14.03	0.831	1.646	0.5745	1.003	1	0
T/P, g=0.3 | long	0.01059	0.03434	0.2067	0.4027	0.2806	0.1221	0.0246	0.9814	0.9784	0
ANN + T/P, g=0.3 | short	0.01175	0.07326	0.2861	0.6458	0.3651	0.1964	0.1031	0.9996	0.9997	0
ANN + T/P, g=0.3 | long	0.007932	0.03983	0.1521	0.3758	0.2992	0.0964	0.02116	0.9142	0.9408	0
T/P/phi, g=0.2 | short	0.01778	0.1016	0.5864	0.7064	0.3844	0.866	0.07675	0.9987	1	0
T/P/phi, g=0.2 | long	0.03421	0.2019	0.9403	0.7536	0.3122	0.4019	0.1072	0.9996	1	0
ANN + T/P/phi, g=0.2 | short	0.01236	0.1245	0.4759	0.5118	0.3779	0.288	0.05025	0.997	0.9999	0
ANN + T/P/phi, g=0.2 | long	0.03442	0.2804	1.006	0.4794	0.4865	0.4599	0.08129	0.9984	1	0

ANN: common n=18, complete short=18/18, long=18/18; ICs=[4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]
P-only: common n=18, complete short=18/18, long=18/18; ICs=[4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]
ANN + P-only: common n=18, complete short=18/18, long=18/18; ICs=[4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]
T/P: common n=18, complete short=18/18, long=18/18; ICs=[4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]
ANN + T/P: common n=18, complete short=18/18, long=18/18; ICs=[4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]
T/P/phi: common n=18, complete short=18/18, long=18/18; ICs=[4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]
ANN + T/P/phi: common n=18, complete short=18/18, long=18/18; ICs=[4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88]

Short: 10,000 updates. Long: scalar training-MSE plateau or 150 s of cumulative new active work/model; plateau 0/240.
Gamma=0.1 ANN/T-P origins were rebuilt because saved Adam states were unavailable; bootstrap counts within 150 s. Other 220 origins preserve exact Adam/RNG.
Incomplete profiles have no full-horizon score. Pairwise cohorts may differ between teams; these rows do not establish an overall architecture ranking or all-species optimum.